

CHE - Chemical Engineering

Chemical Engineering: CHE

Lower-Division Courses

CHE X02. Introduction to Chemical Engineering.

Introduction to chemical engineering, including problem solving and study skills.
Opportunities and responsibilities of a career in chemical engineering.

CHE X03. Introduction to Undergraduate Research.

Introduction to undergraduate research including the structure of academic research, how to access research resources on campus, and foundational skills to succeed in a research environment

CHE X10. Introduction to Computing.

Computer programming focusing on basics of computing, high-level programming environments, and spreadsheets, with application to chemical engineering.

CHE X11. Engineering Sustainable Technologies.

Flows of materials and energy in engineering environments at local, regional, and global scales, and the interaction of those anthropogenic flows with natural cycles of materials and energy. Discusses biogeochemical flows (grand cycles) and anthropogenic material flows at the national level, in industrial sectors, and for consumer products.

CHE X17. Introduction to Chemical Engineering Analysis.

Principles and applications of material and energy balances in process analysis.

CHE X18C. Peer Mentorship.

Best practices for delivery of tutoring and/or grading for undergraduate chemical engineering majors.

CHE X19. Transport Phenomena.

Basic study of momentum, energy, and mass transport. Includes viscous and turbulent flow, heat transfer, and mass diffusion.

CHE X19S. Topics in Chemical Engineering.

Used to record credit the student earns while enrolled at another institution in a program administered by the University's Study Abroad Office. Credit is recorded as assigned by the study abroad adviser in the McKetta Department of Chemical Engineering.

University credit is awarded for work in an exchange program; it may be counted as coursework taken in residence. Transfer credit is awarded for work in an affiliated studies program.

CHE X73K. Process Design and Operations.

Process design, economics, and safety; design projects representing a variety of industries and products.

Upper-Division Courses**CHE X22. Thermodynamics.**

Introductory course in thermodynamics with special reference to chemical process applications: basic laws, thermodynamic properties of single component systems, expansion and compression of fluids, heat engines, multicomponent systems, physical equilibrium, chemical equilibrium.

CHE X22M. Molecular Thermodynamics.

Statistical and molecular concepts, especially the role of the microscopic chemical potential.

CHE X23. Chemical Engineering for Micro- and Nanofabrication.

Definition and description of the terminology and processes of microelectronics, including semiconductor facilities and chemical processes for integrated circuit manufacture, with an emphasis on unit processes; the major unit process, including

thin-film metals and dielectrics deposition and etching, silicon oxidation and etching, ion implantation, diffusion, lithography, planarization, and process control; and an overview of promising nanopatterning and nanofabrication techniques, such as particle-beam imaging, nanoimprint, and near-field probe imaging, implantation, diffusion, lithography, planarization, and process control.

CHE X29S. Topics in Chemical Engineering.

Used to record credit the student earns while enrolled at another institution in a program administered by the University's Study Abroad Office. Credit is recorded as assigned by the study abroad adviser in the McKetta Department of Chemical Engineering.

University credit is awarded for work in an exchange program; it may be counted as coursework taken in residence. Transfer credit is awarded for work in an affiliated studies program.

CHE X33T. Engineering Communication.

Advanced technical communication skills, with emphasis on writing strategies for technical documents, oral presentations, and visual aids.

CHE X37. Quantitative Analysis of Cellular and Molecular Biology.

Analyzes biological systems from stoichiometric, thermodynamic, and kinetic perspectives. Case studies will illustrate how these principles are used to understand disease, control cellular behavior, and design protein-based therapeutics.

CHE X38. Biochemical Engineering.

Introduction to basic biological processes including transcription, translation, protein/enzyme function, cellular energetics, protein secretion and modifications.

Application of quantitative engineering principles to the analysis of biological processes, including thermodynamics, kinetics and stoichiometry.

CHE X39. Introduction to Biochemical Engineering.

Microorganisms in chemical and biochemical syntheses; genetic manipulation of cells by classical and recombinant DNA techniques. Enzyme technology; design of bioreactors and microbial fermentations; separations of biological products.

CHE X39P. Introduction to Biological Physics.

Diffusion, dissipation, and driving forces in cellular processes. Locomotion of bacteria, basic modeling of biomolecular folding and binding events, osmotic flows, and self-assembly in cells.

CHE X39T. Cell and Tissue Engineering.

Introduction to biomedical research in tissue engineering. Includes case studies of tissues and organs of the body, physiology and biology of tissue, pathologies of tissue, current clinical treatments, the role of engineers in development of new technologies to diagnose and treat pathologies, quantitative cellular and molecular techniques, and applications of synthetic and natural biomaterials.

CHE X41. Design for Environment.

Overview of environmental assessment tools for chemical processes and products, including life cycle and risk assessments. Overview of design tools for improving environmental performance of chemical processes, including unit operations and flowsheet analysis methods.

CHE X42. Chemical Engineering Economics and Business Analysis.

Study of the economic decisions faced by chemical engineers. Discounted cash flow techniques. Personal finance, managerial economics, and other special topics.

CHE X43. Molecular Simulation of Materials.

Introduction to basic molecular simulation techniques including molecular mechanics, molecular dynamics, and Monte Carlo method. Understanding of principles underlying these techniques, and how these techniques can be used to study the physical and chemical properties and behavior of materials at the molecular level. More advanced topics include molecular simulations in various ensembles (NVE, NVT, NPT, grand

canonical), free energy computations, controlling dynamics, and association-bias Monte Carlo method. Elementary knowledge of physical chemistry, classical mechanics, and statistical thermodynamics is assumed.

CHE X44. Introduction to Therapeutics.

CHE X46F. Atmospheric Chemistry and Physics.

Examine the sources of air pollutants and the chemical reactions and physical processes that affect them. Explore tropospheric chemistry and the microphysics, chemistry, and thermodynamics of atmospheric nanoparticles. Discuss aerosol transmission of viruses and mitigation of that transmission, as well as recent advances in understanding air pollution and its health effects.

CHE X48. Numerical Methods in Chemical Engineering and Problem Solving.

Numerical solutions to algebraic and differential equations; numerical methods to integration, interpolation, and regression analysis, with application to chemical engineering.

CHE X49D. Dynamics in Ecology and the Environment.

Examine the Earth's natural processes, populations, and systems. Explore the Earth's dynamic systems and how they change with time in response to a variety of disturbances. Use mathematical models to aid in understanding these dynamic systems and to estimate the impact of manmade disturbances. Discuss current environmental and ecological problems.

CHE X50. Chemical Engineering Materials.

Metallic, ceramic, polymeric, and composite materials. Crystal structures, phase diagrams, diffusion, and mechanical properties. Emphasis on structure-property-processing relationships.

CHE X53K. Applied Statistics.

Statistical methods such as data exploration and summary, least-squares fitting, probability and probability distributions, statistical inference and hypothesis testing,

analysis of variance, design of experiments, statistical quality control, and use of professional statistical software.

CHE X53M. Measurement, Control, and Data Analysis Laboratory.

Laboratory safety; measurement and statistical analysis of transport process variables like temperature, pressure, and flow rate; computer data acquisition; feedback control; statistical process control and design of experiments; and production of professional-level lab reports.

CHE X54. Transport Processes.

Design and analysis of heat exchangers, fluid-flow systems and equipment, and interphase-contact devices.

CHE X55. Introduction to Polymers.

Synthesis, structural characterization, physical properties, and applications of polymers.

CHE X56. Optimization: Theory and Practice.

Techniques of optimization, including formulation of optimization problems, one-dimensional search techniques, analytical methods, and n-dimensional search techniques; application of methods to process-industry problems.

CHE X57. Technology and Its Impact on the Environment.

Study of sources and fates of environmental pollutants; environmental quality standards--their measurement and regulation; and pollution control design procedures.

CHE X59. Energy Technology and Policy.

Technology and policy related to energy supply and demand, oil and gas production, coal utilization, hydrogen production, fuel cells, transportation, nuclear power, solar and wind energy, biomass utilization, energy conservation, and climate change.

CHE X60. Process Control.

Analysis of process dynamics and methods for the design of automatic control systems for chemical process plants.

CHE X63. Separation Processes and Mass Transfer.

Design and analysis of equilibrium and mass transfer based on separations such as absorption, chromatography, crystallization, distillation, extraction, and membrane-based processes.

CHE X64. Chemical Engineering Process and Projects Laboratory.

Experimental studies of unit operations. Laboratory safety. Statistical data analysis. Written and oral reports.

CHE X64S. Chemical Process Safety.

Emphasizes quantitative engineering analysis based on the application of mass and energy balances, fluid mechanics of incompressible, compressible and two-phase fluids, heat transfer and conservation of energy, diffusion and dispersion under highly variable conditions, reaction kinetics, and process control. Subjects include various probabilistic and statistical methods to characterize accident and loss performance, techniques for process hazard analysis, risk assessment, and accident investigations.

CHE X72. Chemical Reactor Analysis and Design.

Planning and design of commercial chemical and biochemical reaction systems for producing fuels, polymers, specialty and consumer products, pharmaceuticals, solid-state devices, and other products.

CHE X73K. Process Design and Operations.

Process design, economics, and safety; design projects representing a variety of industries and products.

CHE X76K. Process Evaluation and Quality Control.

Use of statistical techniques to evaluate, compare, and optimize processes. Design of experiments for improved product quality control.

CHE X77C. Undergraduate Research Practicum.

Discuss best practices for participation in research for undergraduate chemical engineering majors.

CHE X77D. Undergraduate Research Practicum.

Discuss best practices for participation in research for undergraduate chemical engineering majors.

CHE X77K. Undergraduate Research Project.

Recommended for students considering graduate study. Topic to be selected in conjunction with individual chemical engineering faculty member, with approval by the department chair. A final written report is required.

CHE X77L. Undergraduate Research Project.

Subject matter to be selected in conjunction with individual chemical engineering faculty member, with approval by the department chair. A final written report is required.

CHE X79. Topics in Chemical Engineering.

Special topics of current interest.

CHE X79.2. Nanomaterials Chemistry and Engineering.

CHE X79.3. Greenhouse Gas Control Technology.

CHE X79.43. Entrepreneurship.

CHE X79.6. Materials Physics.

CHE X79.64. From Data to Decisions.

CHE X79.8. Advanced Nanotechnology and Innovation for Beginners.

CHE X79H. Undergraduate Honors Thesis.

Research performed during two consecutive semesters under the supervision of a chemical engineering faculty member; topics are selected jointly by the student and the faculty member with approval by the department chair. The student makes two oral presentations and writes a thesis.

CHE X79I. Undergraduate Research Internship.

Participate in an internship or career experience program.

Graduate Courses**CHE X73K. Process Design and Operations.**

Process design, economics, and safety; design projects representing a variety of industries and products.

CHE X80C. Laboratory Safety.

Safe laboratory practice. Training in use of fire extinguishers and first aid. Case studies of laboratory accidents.

CHE X80D. Research and Achievements Seminar.

Seminar series covering recent research and technical achievements in Chemical Engineering.

CHE X81N. Fluid Flow and Heat Transfer.

Advanced treatment of fluid flow and heat transfer problems in chemical engineering.

CHE X81P. Advanced Analysis for Chemical Engineers.

Applications of mathematical methods to chemical engineering problems, with emphasis on differential equations, linear analysis and matrices, and real analysis and complex variables.

CHE X81Q. Quantitative Analysis of Cellular and Molecular Biology.

Analyzes biological systems from stoichiometric, thermodynamic, and kinetic perspectives. Case studies illustrate how these principles are used to understand disease, control cellular behavior, and design protein-based therapeutics.

CHE X84E. Electrochemistry and Electrochemical Engineering.

Fundamental principles of electrochemistry combined with mass transfer including applications in analytical chemistry as well as industrial electrochemistry.

CHE X84K. Chemical Kinetics and Surface Chemistry.

Application of chemical reaction kinetics to the prediction and determination of reaction rates and reaction selectivity.

CHE X84S. Current and Emerging Trends in Chemical Engineering Research.

Overview of current and emerging trends in chemical engineering research with frequent guest presentations by leading scholarly and industrial researchers in the discipline.

CHE X84T. Topics in Chemical Engineering.

CHE X85. Research.

CHE X85C. Research.

CHE X85M. Surface Phenomena.

Liquid/fluid interfaces including equilibrium and nonequilibrium phenomena. Topics covered include capillarity, thermodynamics, surface rheology, and streaming potentials.

CHE X85P. Optimization: Theory and Practice.

Techniques of optimization, including formulation of optimization problems, one-dimensional search techniques, analytical methods, and n-dimensional search techniques; application of methods to process-industry problems.

CHE X86K. Theory of X-Ray Diffraction.

Application of basic diffraction theory to polycrystalline and single crystal materials.

CHE X86L. Laboratory Experiments in X-Ray Diffraction.

Application of X-ray diffraction techniques to the examination of polycrystalline and single crystal materials.

CHE X87K. Advanced Thermodynamics.

Applications of thermodynamics to chemical engineering processes.

CHE X87M. Mass Transfer.

Advanced treatment of diffusional mass transfer operations in chemical engineering.

CHE X88K. Separations Processes.

Advanced treatment of modern chemical engineering separations processes.

CHE X91. Elements of Modern Control Theory.

Introduction to fundamentals of dynamic optimization and system theory; applications to engineering processes.

CHE X91J. Reaction Kinetics.**CHE X91M. Mathematical Modeling of Engineered Systems.**

Introduction to fundamentals of dynamic system modeling and associated numerical solution methods with applications to engineering systems.

CHE X91S. Molecular Simulation of Materials.

Introduction to basic molecular simulation techniques including molecular mechanics, molecular dynamics, and Monte Carlo method. Understanding of principles underlying these techniques, and how these techniques can be used to study the physical and chemical properties and behavior of materials at the molecular level. More advanced subjects include molecular simulations in various ensembles (NVE, NVT, NPT, grand canonical), free energy computations, controlling dynamics, and association-bias Monte Carlo method. Elementary knowledge of physical chemistry, classical mechanics, and statistical thermodynamics is assumed.

CHE X92. Polymer Science.

Details of polymerization mechanisms; structure-property relationships, fundamentals of processing, and characterization of high polymers.

CHE X92P. Introduction to Polymer Materials Science.

Synthesis, structural characterization, physical properties, and applications of polymers.

CHE X95C. Chemical Processes for Microelectronics.

Introduction to the chemical processes and the manufacturing operations used in microelectronics device fabrication.

CHE X95E. Polymer Science and Engineering Laboratory.

Training in the preparation and instrumental characterization of polymers, blends, and compounds.

CHE X95G. Chemical Engineering Economics and Business Analysis.

Study of the economic decisions faced by chemical engineers. Discounted cash flow techniques. Personal finance, managerial economics, and other special topics.

CHE X95J. Product and Process Development.

Product and process innovation in the process industries; screening criteria; needs-requirements research; evaluation.

CHE X95K. Design for Environment.

Overview of environmental assessment tools for chemical processes and products, including life cycle and risk assessments. Overview of design tools for improving environmental performance of chemical processes, including unit operations and flow sheet analysis methods.

CHE X97M. Graduate Research Internship.

Research associated with enrollment in the Graduate Research Internship Program (GRIP).

CHE X98. Thesis.

CHE X98R. Master's Report.

Preparation of a report to fulfill the requirement for the master's degree under the report option.

CHE X98T. Supervised Teaching in Chemical Engineering.

Teaching under the close supervision of the instructor for one to four semesters; weekly group meetings; individual consultation; reports.

CHE X99W. Dissertation.

Professional Courses

CHE X73K. Process Design and Operations.

Process design, economics, and safety; design projects representing a variety of industries and products.